

Kristijonas Mikas Silius

Vilnius

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SUMMARY

Data scientist and physics researcher with frontier deep tech focus and CERN background. Master's graduate in Theoretical Physics and Astrophysics. Experienced in developing language models and graph neural networks for high energy physics, production recommendation systems, and AWS infrastructure.

SKILLS

Python, PyTorch, Machine Learning, Data Science, SQL, Data Visualization, Pandas, AWS, Graph Neural Networks, LLMs, Apache Spark, C++, Git, Tensorflow, Keras, Linux OS, MLOps, Bash, LaTeX, ROOT

WORK EXPERIENCE

Telia

Data Scientist | Jul 2025 - Present

Vilnius, Lithuania

- Took technical ownership of recommendation system, deploying and maintaining recommendation models in production on AWS, owning end-to-end lifecycle from data pipeline to model serving.
- Designed pipeline solutions to integrate data from multiple third-party sources with conflicting schemas and quality issues.
- Built A/B testing infrastructure enabling parallel model experimentation and iteration on production recommendation algorithms.

Vilnius University

AI Researcher (Part-time) | Mar 2025 - Jan 2026

Vilnius, Lithuania

- Formalized Master's thesis research into publishable research project on machine learning for high energy physics particle identification.
- Developed fine-tuned language models (HEP-LLM) for domain-specific high energy physics applications using instruction tuning and LoRA with model comparison evaluation framework.
- Implemented automated ArXiv data pipeline for HEP paper collection and processing with quality filtering and text extraction.

CERN

Associated Researcher | Jun 2022 - Jan 2026

Geneva, Switzerland (remote)

- Conducted research on graph neural networks (ParticleNet) for particle identification in CMS experiment at LHC.
- Developed methodology for validating machine learning models on real experimental data using template fitting.
- Collaborated in CMS ttH multilepton working group, presented results at CERN Higgs workshop.
- Optimized deep learning models for Higgs pT regression and collision data simulation for ML model training.

Telia

Data Scientist (Internship) | Jul 2024 - Sep 2024

Vilnius, Lithuania

- Developed features to evaluate TV recommendation model performance using technical and business metrics.
- Implemented quality checks during model training and optimized hyperparameters for improved model performance.

Vilnius University

Senior Laboratory Assistant | Sep 2023 - Apr 2024

Vilnius, Lithuania

- Performed precision studies on particle collision data at the Experimental Nuclear and Particle Physics Center.
- Simulated collision processes for ANN training and optimized neural network architecture.

EXTRACURRICULARS

Masterclass in particle physics — Organizer

- Organized and moderated activities for high school students in data analytics basics used in CERN LHC experiments.

Open Readings — Speaker

- Presented at the 66th International Conference for Students of Physics and Natural Sciences.
- Awarded for best poster presentation on the calibration of CROC pixel detectors.

Hadron therapy masterclass — Organizer

- Organized practical tasks for cancer treatment radiation therapy planning using `matRad` with various irradiation methods.

EDUCATION

Vilnius University

Master of Theoretical Physics and Astrophysics | Sep 2023 - Jun 2025

GPA: 9.6/10

Vilnius University

Bachelor of Physics | Sep 2019 - Jun 2023

GPA: 9.1/10